

SYSTEM DEVELOPMENT

Introduction

System development involves identifying business requirements and developing information systems that will effectively help to support the day-to-day operations & decision-making processes in an organization.

Definition of a System

- ❖ A **System** is a set of organized components which interact in a given environment and within a specified boundary to achieve collective goals & objectives that are emerging.
- ❖ A **System** is a set of items, equipments, procedures, processes, techniques, programs & people working jointly with an aim of achieving common goals.

Examples of systems are:

- Education (school) system.
- Transport system.
- Bicycle system.
- Banking system, etc.
- ✓ A system is made of various components. Once the components come together, they become interrelated to each other and generate new goals and objectives, and as such, a system breaks down when any of its components is removed.
E.g., a Bicycle system has all the components working together to provide motion when ridden. The individual components cannot provide these services to a rider when on their own.
- ✓ A system does some useful job; hence, it should be active & efficient, e.g., a Banking system deals with money.

Description of a System

A system can be described as either being **Soft** or **Hard**.

Soft systems:

These are usually the human activity systems.

They are described as soft because of 3 main reasons;

1. Their boundaries keep on changing.
2. Their goals & objectives usually conflict, and may not be captured clearly at any one given time. This is because; they are based on human factors such as *attitudes & preferences*.
3. It is difficult to clearly define their exact measures of performance.

Examples of soft systems:

- i). **A Political system:** - it is very difficult to come up with a system that will predict the political mood in a country over a given period of time.
- ii). **A sales tracking & prediction system in an organization:** - sales in an organization depend on human factors such as attitude in the market place.

Hard systems:

These are systems whose goals & objectives are clearly defined, and the outcomes from their processes are predictable and can be modeled accurately.

Hard systems are based on proven scientific laws such as mathematical formulas or Engineering solutions.

Example of a hard system:

- i). **Stock management system in a supermarket:** - it is possible to know exactly the stock levels, cost, selling price, and to predict accurately the profit if all the stock is sold.

Note. A good system should have features of both soft & hard systems, e.g., a stock management system should be able to show when the demand for a certain item rises so that a decision can be made on when to buy more stock. Similarly, new demand is driven by soft aspects in people's lives such as attitude & seasons.

CHARACTERISTICS (FEATURES) OF A SYSTEM

All systems have some common characteristics. Some of these characteristics are:

1). Holistic thinking

A system contains a set of interacting elements. However, in holistic thinking, a system is considered as a whole unit.

Note. The concept of a system emerged from early psychologists who believed that the mind was a whole unit, rather than a collection of psychological parts.

The various components that make up a system may be simple in nature, but when combined, they create something complex whose overall goals are more sophisticated than those of the individual components.

2). Purpose

A system must be designed to achieve a specific predetermined objective, e.g., one main objective of a *school system* is to enable the students to excel in national examinations.

The objectives that a system is supposed to achieve will enable the system developers measure the performance of the system during its operation.

3). System boundaries and environment

Each system is required to operate within a specific framework or limits. The space within which the components of a system operate is known as its **boundary**. Outside this boundary is the **environment**, from which inputs are received & to which outputs are communicated.

Entities that fall outside the boundary but interact with the system are called **external entities**, and they form part of the system environment. External entities provide the inputs & also receive the outputs from the system.

e.g., the external entities to a school system may include; Parents, various suppliers, and the society.

Therefore, a system operates within specified boundaries, and interacts with other systems.

4). Sub-systems

Each system is made up of different components (or other systems) that communicate with each other. These systems are described as **Sub-systems**.

This means that, a system does not exist alone, but it is composed of subsystems, which are also made up of other subsystems.

Example;

The *Classroom system* is part of a *School system*, which is a subsystem of the *Ministry of Education*. The Ministry of Education is part of the *Government*, while the Government is part of the *Global system*.

Therefore, every system is a component of a larger system.

5). Inputs & outputs

A system communicates with its environment by receiving inputs & giving outputs.

Example;

A manufacturing firm can be considered as a system that gets inputs in form of raw materials from the environment and transforms them into finished products (outputs), which are released into the environment.

6). Process

A system will usually transform data from one state to another.

Usually, after the raw data is collected & prepared into a form suitable for input, it is then manipulated into information using the given procedures or instructions. The processing may be manual, clerical, electro-mechanical or automatic to obtain the information.

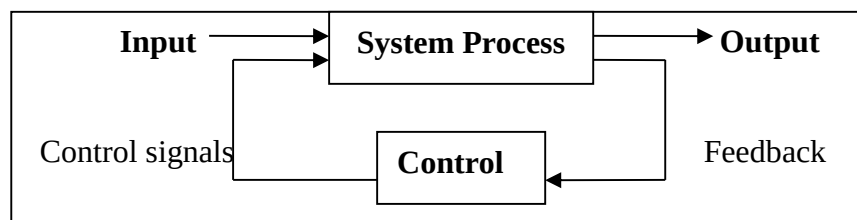
7). System control

A system has some controls that help it not to operate beyond its boundaries. **Control** is the method by which a system adapts to changes in the environment in order to give the expected output or to perform to the expected level.

Control is normally achieved through feedback. **Feedback** is a check within a system, which ensures that the objectives of the system are achieved. They assist the system by monitoring the environment in which it operates in order to find out any deviation. If any deviation is detected, then the appropriate steps are taken to rectify this error.

The feedback may involve having the outputs from the process of the system being fed back to the control mechanism. The control mechanism will then adjust control signals that are fed to the process, which then ensures that the output meets the set expectations.

The figure below shows a system that has feedback to the control function.



Example;

A motor vehicle manufacturing company is expected to produce several vehicles per day. If the demand increases, the feedback will show that the company is underperforming. Control signals can then be issued to speed up the movement of units on the Assembly line so as to increase production.

8). System entropy (decay)

A system slowly becomes useless to the user either due to improvement in technology, new management policies or change in user requirements.

Therefore, a system must be reviewed with the aim of improving it or to developing a new one.

9). A system must give priority to the objectives of the organization as a whole as compared to the objectives of a subsystem.

Open and Closed systems

A system can be described as being **open** or **closed**.

Open systems:

An Open system is that which interacts and communicates with its environment constantly. It receives inputs from & gives output to the environment.

Examples of open systems;

- ◆ Business organizations.
- ◆ Information systems.
- Open systems normally adapt to changes in the environment, e.g., for any business organization to exist, it must be able to adapt to the changing market prices, competition, etc.
- An open system has an unlimited scope when providing the services of the organization.
- In an open system, relevant variables keep on interacting.
- An open system is generally flexible & abstract.

Closed systems:

A Closed system is that which does not interact or communicate with its environment. It does not communicate to or receive communication from its environment.

- Closed systems do not receive inputs or give output to their environment.
- A closed system can correct or control itself; hence, it does not obtain modification from its environment.
- A closed system has a limited scope when providing the services of the organization.
- The variables in a closed system are self-contained.
- A closed system is rigid & mathematical.

Example of a closed system;

- ◆ A computer Program. It accepts previously defined inputs, processes them & gives previously defined outputs.

Revision Questions.

1. What do you mean by a System?
2. Differentiate between soft systems and hard systems.
3. List five characteristics of a system.
4. Explain the term system boundary.
5.
 - a). What is system Control?
 - b). Why do we need feedback in a system?
 - c). Draw a well-labelled diagram that shows a system that is controlled through feedback.
6. Distinguish between open and closed systems.

INFORMATION SYSTEMS

Definition of an Information system:

An **Information system** is an arrangement of people, data processes & information that work together to support and improve the day-to-day operations in a business and the decision-making process.

Generally, the Information system of an organization is the complete apparatus for handling all aspects of information within an organization. It includes people, procedures, technological, and other resources that collect, transform & disseminate information in the organization.

Purposes of an Information system:

The main purposes of an Information system in an organization are: -

1. To support information processing tasks such as data collection, processing & communication.
2. Help in decision making by collecting data, analyzing it, and generating reports.
The process whereby a computer-based information system is used to capture operational data, analyze it, and generate reports that can be used to support the decision making process in an organization is referred to as *Online analytical processing*.
3. Supports sharing of information between departments/users in a given organization. The departments can share the same electronic information stored in central database.

Circumstances that necessitate the development of new Information systems:

The following are some of the circumstances that bring about the need to develop new information systems:

- 1). *New opportunities*: - a chance to improve the quality of internal processes and service delivery in the organization may arise.
- 2). Invention of new systems which are more successful than the existing ones.
- 3). *Problems*: - the user may encounter some difficulties in the operations of the existing system, which prevent the organization from meeting its goals.
- 4). The management may identify an area of poor performance, which increases the level of indirect expenses.
- 5). *Directives*: - these are requirements imposed by the management, government, or external influences.

System Analysis and Design

System Analysis:

- ◆ *System analysis* is the process of examining an activity, procedure, technique or organization to determine what must be accomplished and how best the operations may be accomplished in order to achieve the predetermined objectives.
- ◆ The process of collecting & analyzing facts that relate to an existing situation, diagnosing problems, and using the facts gathered to design & implement an effective computerized system.

System Design:

System design is the activity that involves identifying possible solutions to a problem, and then deciding on the most appropriate system to solve the problem.

System design is concerned with the design of a computerized application based on the facts disclosed during the Analysis stage.

In system design, the nature & contents of inputs, files & outputs are formulated and described in order to show how they are connected by processing procedures, and for the purpose of developing a new (or, an improved) system.

Information system Analyst:

An *Information system Analyst* is a person who identifies the problems & needs of an organization, then designs & develops algorithms and procedures on how to solve these problems on a computer.

The Analyst uses scientific techniques so as to determine where & how improvements can be made in order to meet objectives in a more efficient, efficient, and economical manner.

Roles of an Information system analyst

1. Reviews the existing system & makes recommendations on how to improve or implement an alternative system.
2. Works hand in hand with programmers to construct a computerized system.
3. Coordinates the training of new system users and owners.
4. He is the overall project manager of the information system being implemented. Some of his project management duties include: assuring quality, keeping within schedule & budgeting.

Revision Questions.

1. Define the following terms:
 - i). Information system.
 - ii). Online analytical processing.
2. State and explain three purposes of Information systems in organizations.
3. Highlight three circumstances that necessitate the development of new information systems.
4. List down four roles of an Information system analyst.

THEORIES OF SYSTEM DEVELOPMENT

There are 3 main theories or methods used in system development.

1. Traditional approach.
2. Rapid Application Development (RAD).
3. Structured approach.

Traditional approach.

In the Traditional approach, there is no formal documented methodology to be followed by all system developers in the organization.

The method relies mostly on the skills & experience of the individual members carrying out the project development.

Disadvantage of traditional approach.

- i). The structure of the old system is not changed in anyway; hence, the weaknesses of the old system are not corrected, and are carried forward to the new system.

Example;

In a Bank, a manual system is characterized by long queues & poor controls. If the Traditional approach is used, each Cashier will simply be given a computer. The long queues might remain and lack of controls increase because no value was added to the old information system.

Rapid Application Development (RAD).

This method heavily relies on Information Technology (computers). This is because; there is need for businesses or organizations to develop & implement information systems quickly enough for them to maintain a competitive advantage in the market.

Advantage of RAD.

- Ensures faster development of information systems, and as a result, increase production of an organization.

Disadvantage of RAD.

- The working system may be weak due to quick development, i.e., a system may be working well but may not have the necessary inbuilt security mechanisms.

Approaches used in Rapid Application Development.

There are 3 different approaches/techniques used in Rapid Application Development;

- (a). Prototyping.
- (b). Small team With Advanced Tools (SWAT).
- (c). Joint Application Development (JAD).

Prototyping

A **Prototype** is a small working model, which is developed to test ideas & assumptions about the new system.

Like any computer-based system, a prototype consists of working software that accepts input, performs calculations, produces printed or displayed information, or performs other meaningful activities. The design & the information produced by the system are tested & evaluated by the users.

Note. Use of prototypes makes it possible for system developers to quickly capture user requirements by designing system interfaces in the presence of the user.

Structured approach

In structured approach, there is a defined set of stages that should be followed when developing a system. Each stage is well documented and specifies the activities to be carried out by the system analyst and his team while developing a system.

Revision Questions.

1. Explain three system development theories and methods.
2. (a) State one disadvantage of the Traditional approach in system development.
(b) State the main advantage of Rapid Application Development method.

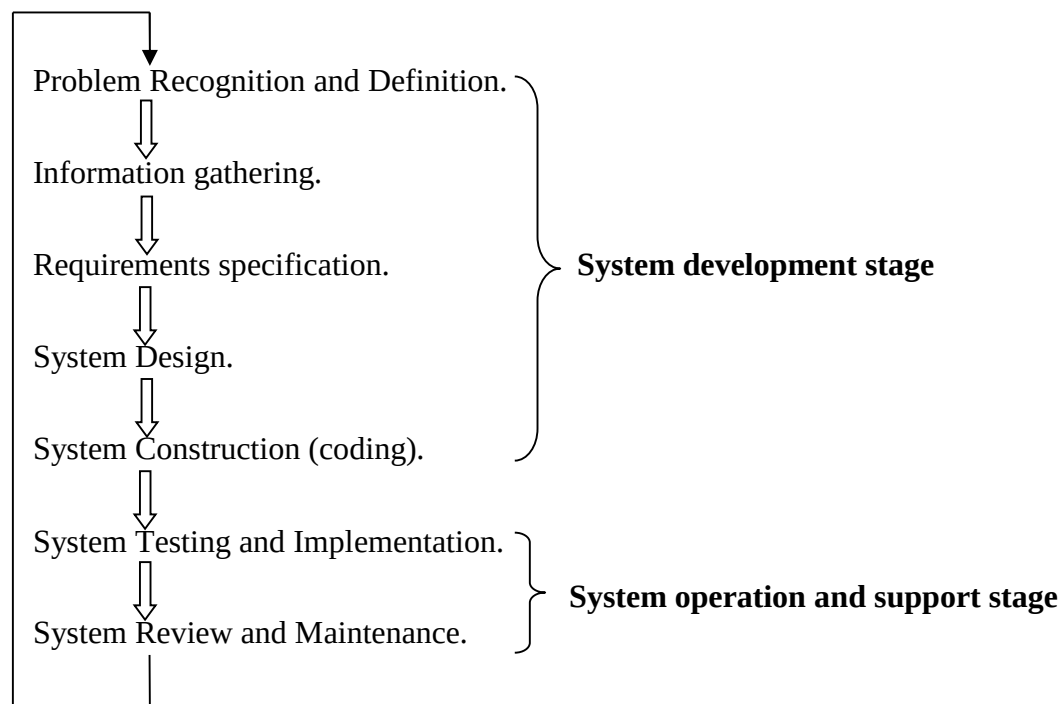
SYSTEM DEVELOPMENT LIFE CYCLE

The stages of developing a system are called the **System development life cycle (SDLC)**.

The 7 main stages in system development include:

1. Problem recognition and definition.
2. Information gathering.
3. Requirements specification.
4. System design.
5. System construction (coding).
6. System testing and implementation.
7. System review and maintenance.

The following diagram represents the 7 stages, which must be followed in the system development life cycle: -



The life cycle of an information system is divided into 2 major parts: -

- a). The development stage.
- b). The operation and support stage.

Note. Each stage serves a role in the problem-solving process, and therefore they must be followed systematically.

Problem recognition

Problem recognition is done during the **Preliminary investigation**. A *preliminary investigation* is carried to find out if really there is need for change.

During the problem recognition stage, the system analyst seeks to answer two questions:

1. Is the proposed project worth looking at?
2. Is the project worth pursuing?

After this, the system analyst then defines the scope of the project and tries to establish the limitations (risks involved), the budget (i.e., cost, resources/manpower involved) & time involved.

Some of the most common limitations are:

- Lack of finance, and lack of appropriate technology (expertise) to develop the system.

Problem definition (Problem analysis)

Problem definition is the process of identifying & understanding the problem, and finding out any limitations that may limit the solution.

At this stage, the system analyst is required to find out much about the existing system (whether manual or computerised) in order to come up with a good & relevant proposal for the new system.

A special study called a *feasibility study* is carried out. A **Feasibility study** is a study carried out to establish the costs & benefits of the proposed new system.

The study tries to;

1. Justify the new system in terms of the capital to be employed, equipments required, personnel, and the procedures necessary for the new system.
2. Determine whether the existing system, be it manual, mechanical or computerized, is adequate, or it should be modified, updated or replaced.

The feasibility of a system is accessed in 4 ways: -

1). **Technical feasibility.**

This tries to establish whether the existing technology can be developed (upgraded) or is sufficient to support the new system.

It also tries to find out whether the staff has relevant technical skills to develop & use the new system.

2). **Economic feasibility.**

Economic feasibility study tries to establish whether developing the new system is cost effective by comparing all the costs & benefits of the proposed system.

I.e., it tries to find out whether the expected benefits will exceed the costs of developing & operating the proposed system.

3). **Operational feasibility.**

Operational feasibility is concerned with the operation of the office. It establishes whether the management, employees, customers, suppliers & other users are happy, willing and able to operate, use and support the proposed new system.

For example;

When carrying out operational feasibility study, the analyst tries to:

- Ask the users whether they accept the new system?
- Find out whether the staff has the necessary skills or manpower? If they don't have the necessary skills, how are they going to acquire them? Do they require training or not?
- Look at the kind of job assignments that will be affected.
- Look at how the firm will be reorganized to accommodate the new system, e.g., office space, comfort, etc.

4). **Schedule feasibility.**

It establishes whether developing of the proposed system will be accomplished within the available time.

Note. This cost-benefit analysis study will then indicate whether the proposed system is viable or not. Otherwise, a new system should only be developed if its benefits are more than its costs.

After the feasibility study, a feasibility study report is produced, which outlines the following:

1. Recommendations on whether to continue or abandon the project.
2. The performance of the new system in relation to the existing system.
3. Limitations & benefits expected.
4. The development plan for the new system, etc.

The recommendations contained in the report are carefully evaluated by the personnel involved in the system study, i.e., the *management, user departments, steering committee, finance department*, etc who will then decide on whether to commence a detailed investigation or not.

Revision Questions.

1. Define the term *System development life cycle*.
2. Outline the seven steps followed in structured system development.
3. What is '*Problem definition*' in the context of system development?
4. Define the term *Feasibility study* as used in system development.

Information gathering (Fact-finding)

After the feasibility study report has been approved by the management, the system analyst then proceeds to identify the techniques that will help the management to gather enough information relating to the starting of the system.

The collection/gathering of all information required to implement a computer system is referred to as **fact-finding**.

Objectives of fact-finding.

The main objectives of fact-finding are:

- To find out what the present system is attempting to do, its scope and objectives.
- To collect information about the input of the present system.
- To find out the volume of input. This directly affects the design of the new system.
- To find out about the files maintained by the present system.
- To find out about the processing carried out by the system.
- To find out how the files are updated & outputs produced.
- To find out what equipments are used.
- To find out about the accuracy checks performed during processing.
- To find about time limitations, if any.
- To find out about the organization structure of the departments and the sections presently carrying out the processing tasks.
- To find out the problems & difficulties presently encountered as the system operates, with special reference to the barriers, duplication, and weakness.
- To ascertain the cost of the present system.

Revision Questions.

1. What do you mean by fact-finding?
2. State the objectives of fact-finding?

Fact-finding techniques

The following are some of the common methods/techniques used to collect data:

1. Document review (study of available documents or records).
2. Use of questionnaires.
3. Interviewing.
4. Automated methods.
5. Use of workshops.

Note. Before the system analyst chooses the most appropriate technique, he/she should compare the merits & demerits of each technique. This will ensure that the technique chosen will be able to meet all the requirements, which will assist the management in achieving its goals.

Document review (study of available documents):

This involves going through all the existing documents/records, which relate to the system being investigated in order to find out information that describe the data & procedures of the current system.

Examples of such documents are: -

- a). *Organizational charts* – they illustrate or describes the formal structure of the parts of the organization relevant to the investigation.
- b). *Reports*, e.g., financial reports, review & evaluation reports.
- c). *Procedure manuals, formal job descriptions & job specifications* – they describe how tasks should be carried out.
- d). Standard & technical manuals.
- e). Strategy & policy documents.
- f). *Forms* used within the system.
- g). *Backup files* that the system maintains.
- h). Card catalogues.
- i). Receipts.

Records inspection involves studying all the manuals maintained in connection to the system being studies. This helps the analyst understand the structure of the organization, its operation, and history.

Advantages of documents review

1. It helps the analyst to gather basic background information about the system.
2. It assists the analyst in designing interview questions & questionnaires.
3. It helps the analyst in deciding whom to interview.

Disadvantages of documents review

1. The documents may not have been modified to reflect the current status of the system, i.e., they may be obsolete. This is because; documents such as organizational charts & procedure manuals may have been documented when the information systems were set up.
2. Human systems are dynamic. This implies that, after sometime, the real system will have changed greatly from the documented one. Therefore, reading such documents will give the analyst a false picture of the system.
3. Documents relating to a system are usually bulky. Therefore, reading all the documents will be time-consuming.
4. The documentation may be poor or incomplete, thus the analyst may not obtain all the information he would require.

Observation:

This method requires the analyst to participate in or watch closely as a person performs some activities for a period of time in order to see for oneself what exactly happens in the system.

In Observation, the analyst asks no questions. Instead, he observes the actions in which he is interested, and records the desired information. This method gives the analyst first hand experience about the problems and exposes him/her to the system requirements.

Advantages of observation

1. Data collected is highly reliable, since the method gives the real picture of the system.
2. Concepts or tasks that are too difficult for non-technical staff to explain in words can be clearly observed.

The analyst is able to see clearly what is being done. He can also identify tasks, which have been omitted or inaccurately described by other fact-finding techniques.

3. Allows the analyst to do some measurements.
4. It is relatively cheap compared to other techniques.

Disadvantages of observation

1. The presence of the analyst may make the person being observed perform differently (or change behaviour) leading to wrong requirements being observed.
2. The work being observed may not involve the level of difficulty or volume normally experienced during that time period.
3. The need to be on the site where the activities are taking place consumes a lot of time.
4. Tasks being observed are subject to various types of interruptions.
5. Some system activities may take place at odd times causing a scheduling inconveniences for the analyst.

Use of Questionnaires:

A *Questionnaire* is a special-purpose document that allows a person to collect information & opinions from respondents.

The method involves sending out forms containing questions with spaces for response to a group of people, and collecting the forms back after they are completed. This method allows the analyst to collect facts from a large number of people while maintaining uniform responses.

Circumstances in which a questionnaire is used for gathering information.

The questionnaires method is used in situations where:

- The information to be gathered is located over widely spread geographical areas.
- A large number of people are to be questioned, and the questions to be asked require short answers, or are limited to Yes/No.
- 100% coverage is not essential.
- Privacy (anonymity) of the respondents is to be maintained.

Note. In a situation where a large population is to be questioned, the analyst may spend a lot of time analyzing the questionnaires. In such cases, a sample of people (who are assumed to represent the overall population), can be given the questionnaires.

Advantages of questionnaires

1. Questions can be answered quickly, since respondents can complete & return the questionnaires at their convenient time.
2. Use of questionnaires gives the respondents privacy; hence, there is likelihood that the information given is sincere & real.
3. Questionnaires provide a relatively cheap means for data being collected from a large number of individuals.
4. Responses can easily be tabulated & analysed quickly.

Disadvantages of questionnaires

1. Good questionnaires are difficult to prepare.
2. Number of respondents is usually low.
3. The respondent may not fully understand the questions because of ambiguity of language; hence, he/she may end up giving wrong responses.
4. No guarantee that an individual will answer or explain on all the questions.
5. No immediate opportunity to clarify an unclear or incomplete answer to any question.
6. Not possible for the analyst to observe & analyse the body language of the respondents.

7. Questionnaires are inflexible, i.e., no opportunity for the analyst to obtain voluntary information from a respondent.

Procedure for developing a questionnaire

- 1). Determine the facts & opinions, which must be collected and from whom you should get them. If a large number of people is involved, then select a sample.
- 2). Depending on the facts required, decide whether to use free-format or fixed-format questions.
- 3). Write the questions & edit them such that they do not offer your personal bias or opinions.
- 4). Test the questions on a sample of respondents, then edit if necessary.
- 5). Duplicate & distribute the questionnaires.

Interviewing:

Interviewing is the process of obtaining information from another party by means of conversation.

Interviews enable the system analyst (who is the Interviewer) to collect information from the affected individual (Interviewee) through face-to-face communication. The Interviewer asks questions and the Interviewee responds with answers.

The analyst should carry out interviews with the relevant stakeholders in order to get views about the current system, and gather information about the requirements for the proposed system.

Interviewing provides facts and also enables the analyst to verify the facts. It also provides an opportunity to meet & overcome any possible user resistance.

A good interview should be planned, and should be carried out at the most appropriate time for the parties involved.

When executing an interview, the following guidelines should be followed:

1. The interviewee must be informed in good time, and the topic of discussion communicated accordingly to allow for adequate preparation.
2. Avoid personal biasness in your questions and perspectives.
3. Be careful about body *proxemics*. **Proxemics** refers to issues related to physical contact such as sitting arrangement or body closeness during an interview.

At the end of the interview, the analyst should read what he/she has recorded to the interviewee for further clarification before the final notes can be taken.

Advantages of interviews

1. An interview provides the analyst with an opportunity to motivate Interviewees to respond freely & openly to questions. This removes the rigidity that exists in questionnaires.
2. It allows the analyst to rephrase/frame questions for each person.
3. It allows the analyst to prompt/urge for more feedback from the Interviewee.
4. It gives the analyst an opportunity to observe the verbal & non-verbal communication such as facial expressions of the interviewee.

Disadvantages of interviews

1. It is difficult to organize interviews. This makes the method time-consuming & costly.
2. Many system analysts are poor interviewers, thus there is a possibility of failure.
3. The interviewee may not fully open up on some issues that may be personal or sensitive.
4. Interviewing may not be practical due to the location of the interviewees.

Automated methods:

Automated data collection is mostly used when actual data is required but difficult to get through interviews, observation, or questionnaires.

Such data may be collected using devices that automatically capture data from the source such as Video cameras, Tape recorders, etc.

Revision Questions.

1. State and explain four methods that can be used to gather information during system development.
2. Identify three disadvantages of the observation method used in fact-finding.
3. Outline three disadvantages of Questionnaires
4. State two advantages and two disadvantages of the interview method when used for information gathering.
5. Explain the concept of proxemics in interviews.
6. (a) Explain the importance of using automated methods in fact finding.
(b) Give one example of automated information gathering technique.

Fact recording:

Fact recording takes place at the same time the analyst is gathering the facts. Facts relating to staff, operations, and processing tasks are recorded.

Fact recording is necessary because; the subsequent stage of system development shall depend on the facts recorded, i.e., the facts recorded will form reference material for the analyst during system design.

The various methods of fact recording are:

- Procedure descriptions (narrative).
- Clerical procedure charts.
- Flow process charts.
- Decision tables.
- Grid charts.

Preparing and presenting the fact-finding report:

After gathering the information/facts, the system analyst must come up with a requirements definition report, which must contain the following details:

1. **Cover letter** addressed to the management and the IT task force written by the person who gathered the facts.
2. **Title page**, which includes; the *Name of the project*, *Name of the analyst*, and the *Date the proposal is submitted*.
3. **Table of contents**.
4. **Executive summary**, which includes recommendations of the system analyst of how the new system is to be implemented. This is because; some people only read the summary to make decisions.
5. **Outline of the system study**, which provides information about all the methods used in the study, who and what was studied.
6. **Detailed results of the study**, which provides details of what the system analyst has found out about the system, e.g., problems, limitations, and opportunities that call for an alternative.
7. **Summary** – a brief statement that reflects the contents of the report. It also stresses on the importance of the project.

This report is then presented to the Management for evaluation and further guidance.

Requirements specification

In requirements specification, the system analyst must come up with detailed requirements for the new system.

The following requirements specifications are considered:

1. Output specification.
2. Input specification.
3. File/data structures.
4. Hardware & software requirements.

Output requirements.

In system development, the output requirements of the new system are considered first. This is because; the main interest from a system is information (output), e.g., the main concern of a library management system is whether the system can generate reports on overdue books, charges of late return, inventory reports, etc.

The output is usually in the form of reports either in the form of hardcopy or softcopy.

The following factors should be considered when designing the output:

- i). *Target audience*: - a user report may show only the transactions to be carried out, while the management would require a summary of the overall performance in the organization.
- ii). *Frequency of report generation* (i.e., the time at which the output is required): - some reports are required daily, others weekly, monthly or annually.
- iii). *Quality and format of information to be generated*.
- iv). *Cost of producing the output*: - the output should be at reasonable cost.
- v). *Mode of output & devices used for output*, e.g., softcopy mode of output is produced through the screen.

After designing the output, it should be approved by the users, the management, and other staff within the organization who are affected by the change.

Input specifications.

Once the system analyst has identified the output requirements for the new computerised system, he/she then identifies the input needed to obtain the relevant information from the system.

The input to the system is necessary because the contents input are used to maintain the master files.

The system analyst should therefore decide on:

1. The contents & volume of input, and whether these contents can grow.
2. The mode of input, the devices of input selected and their suitability.
3. The format & sequence of input.
4. The nature of the system, which determines the frequency of input, e.g., for Batch processing, the frequency would be periodical, and for Real-time systems, it would be on-demand.

After identifying all the inputs, the analyst designs the user interface by designing data entry forms or screens.

When designing the user interface, the following guidelines should be observed:

- a). Objects such as Textboxes, Labels, and Command buttons placed on the forms must be neatly aligned & balanced on the form.
- b). The size of the form should not be too small for the user to read or too big to fit on the screen.
- c). The colour of the interface should not be too bright to avoid hurting the eye.

File requirements specification.

This involves identifying the files required to store data & information in the system.

The system analyst should:

- Identify the number of files that will be needed by the system.
- Determine the structure of each of the files, e.g., will the files allow direct access? Will they be sequential files stored on a magnetic tape?
- Identify the attributes of the records in a file.

An **Attribute** is a unique characteristic of a record for which a data value can be stored in the system database.

Note. These attributes are used when designing tables in a database, and each attribute becomes a field in the table.

For example;

A Books table will have the following attributes/fields: Book ID, ISBN number, Title, Author's name, Year of publication, Date of issue and Date of return.

Book ID	ISBN No.	Title	Author's Name	Year of publication	Date issued	Date of return
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Factors to consider when designing a good file.

- Record key field:** - this is usually an attribute that is unique for each record.
- Data type for each field:** - each field has a data type. In a database, the data type of book titles can be stored as 'Text', while the Date of borrowing a book can be stored as 'Date/Time'.
- Length of each field:** - a field used to store names can be specified to be 30 characters long, while a field used to store numbers/integers can be specified to be 10 characters long.
- Backup and recovery strategies:** - the updated copies of data & information files need to be stored in a different place other than the location of the current system. This ensures that, if the current file gets corrupted, the backed up data can be used to recover/reconstruct the original file.

Hardware & software requirements.

The system analyst should specify all the hardware & software requirements for the new system.

The hardware & software used to develop the system mainly depends on Input, Output & File requirements, e.g., if the system requires data in picture format, then an image capturing device such as a Digital camera or a Scanner must be used.

Some of the factors to consider in hardware & software specification are:

- Cost (price)
- Method of acquisition.
- Reliability & security features.
- Upgradeability.
- Compatibility with the existing resources.
- User friendliness.
- User requirements (user needs)
- Portability, etc

Revision Questions.

- Outline the four areas that are considered during the requirements specification stage.
- Explain at least three factors that should be put into consideration during output design.
- State four factors that may be considered in order to design a good file.
 - Define the term 'Attribute'.
 - Explain why it is important to consider the file backup and recovery strategies during file design.
- Outline the factors that should be considered when sourcing for hardware and software resources required for a new system.

System Design

In the design stage, the analyst must come up with ways of solving the problem.

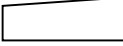
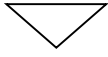
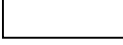
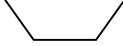
The following are some of the tools used for designing an information system:

- System flowcharts.
- Data flow diagrams.
- Entity relationship models.
- Structured charts.

System flowcharts.

A system flowchart is a tool that can be used for analysing processes. It allows one to break a process down into individual events/activities, and also display these events in a short form showing the sequential or logical relationships between them.

A system flowchart has its own set of symbols. The following are some of the common system flowchart symbols:

	- Start or Stop		- Decision
	- <i>Rectangle with rounded corners</i> : - it represents an event which occurs automatically & usually triggers a chain of other events, e.g., a book lending process is triggered by a student request.		- Disk master file
	- Flow direction		- Disk transaction file
	- Physical flow of goods		- Tape file
	- Data input device (keyboard)		- Offline file
	- <i>Rectangle with sharp corners</i> : - represents a computerized process		- <i>Kite</i> : - represents a sort operation
	- Manual process		- Monitor screen (CRT)
	- Report or document		- On page connector
			- Off page connector
			- Telecommunication link

Designing a system flowchart.

A system flowchart gives a summary of how particular processes are done within the business organization.

The following are some of the important guidelines when designing a system flowchart:

1. Start by writing the title of the flowchart.
2. If possible, start drawing the flowchart with the trigger event.
3. List down the actions taken in their logical order until the process is concluded. Use few words to describe the actions.
4. In case there are many alternatives at the decision stage, follow the most important and continue with it. The less important alternatives can be drawn elsewhere and reference made to them using the On-page or Off-page connectors.

After drawing the system flowchart, other design tools such as **pseudocodes** and **program flowcharts** can be used to extract the processing logic for each module in the system before system construction.

Revision Questions.

1. State two methods that a system analyst may use to design a system.
2. (a) Differentiate between a system flowchart and a program flowchart.
(b) Draw four system flowchart symbols and explain their functions.

System construction

System construction refers to the *coding, installation and testing* of the modules and their components such as outputs, inputs & files.

The purpose of the construction stage is to develop & test a functional system that fulfils the design requirements of a particular organization.

System construction is done by **programmers**.

System construction methods

To construct a system, the programmer can use the following programming techniques:

- 1). Use the high-level structured languages, e.g., *Pascal, COBOL*, etc
- 2). Use the 4th generation languages, e.g., *Visual Basic, Visual COBOL, Delphi Pascal*, etc
- 3). Customise & use a ready-made standard package such as a database software, financial package or enterprise management system.

System testing

After constructing the system, it is tested by entering some test data to find out whether its outputs are as expected.

When the system is newly developed, it can be first tested using *dummy (assumed) data*, while *real/live test data* can be used for normal circumstances to find whether the system can detect & report errors.

System testing is carried out in order to achieve the following aims:

- To test the programs further to detect any errors. This is because; after testing the programs, some errors might have gone unnoticed.
- To find out whether the system meets all requirements specified.
- To establish whether the programs work interactively as a suite of programs.
- To find out whether there is a link between the clerical & computer procedures.

During system testing, the following details should be checked:

- ◆ Files maintained in connection to the information requirements of the system.
- ◆ Input to the system, for the maintenance of the existing files.
- ◆ Processing tasks.
- ◆ Reports generated by the system.
- ◆ Controls incorporated within the system.

NB: System testing is an iterative process, and it ends only when the analyst & the other personnel involved are satisfied that when operational, the system will meet the objectives and the growing demands of the organization.

Revision Questions.

1. (a) Explain system testing.
(b) State the importance of system testing.
(c) Name three groups of people who are involved in system testing.

System implementation

System implementation is the process of putting the new system in day-to-day operating environment for the users to start using it.

A system is put into use after it has been fully tested, well documented, and after training the staff who will be involved in the running of the new system.

In most cases, the implementation phase is faced with various problems. They include:

- Staff problems in the user department, e.g., in case the confidence levels are low, ignorance, etc
- Security aspects, e.g., inadequate controls, are there any standby arrangements in case of failure, etc
- Administrative problems.

The areas to be addressed during system implementation include:

1. File creation & conversion.
2. Staff training.
3. Changeover strategies.

File creation and conversion:

Every time a new system is implemented, the format of data files might change or might require modification.

The files can either be created from scratch or those that were used in the old system can be converted to be used in the new system.

Therefore, **file creation & conversion** involves setting up of the master files that are to be used to support the information requirements of the new system.

The factors to consider during file conversion include:

1. Whether the new system requires a new operating system & hardware.
2. Whether there is need to install new application software, e.g., if the new system will be developed by customising database application software, then there is need to install the software if it is not installed.
3. Whether there is need to create new database files for the new system, e.g., if the files previously used were manual, then electronic ones will have to be created.

Staff training

After designing a new system, all the staff affected by the change should be trained properly on how to use/run the system.

The aims of the training are:

- a). To remove the fear of change in them.
- b). To convince the staff of the user department on the effectiveness & efficiency of the new system.
- c). To enable the staff to cope with the processing tasks of the new system.

The following methods of training can be used depending on the requirements:

- Demonstrations.
- Film shows.
- Group discussions.
- Lectures or seminars.
- Visits.
- Reference from the available documents such as User manuals, etc.

NB: The training should be well planned & the most suitable method that can meet the needs for all the stakeholders selected. Otherwise, if the staffs are not trained properly, the system implementation can fail leading to great loss of company resources.

Changeover strategies:

Changeover is simply how to move from the old system and start using the new system.

The changeover should be planned & effected at the most suitable time for a smooth transition to the new system.

The following are some of the common methods/ways of system changeover:

1. Direct (straight) changeover.
2. Parallel running.
3. Phased changeover.
4. Pilot running.

Direct (straight) changeover:

Straight changeover is a complete replacement of the old system with the new system in one bold move.

In this approach, the old system is stopped & abandoned and the new system starts operating immediately.

This sudden change from old to new can be very inconveniencing in case the new system fails, faces problems, or in a situation where the users have not gained enough confidence to run the new system.

Direct changeover is likely to be used in situations where:

- ✓ The users have a very high level of confidence with the system.
- ✓ The new system bears little resemblance to the old.
- ✓ The system is relatively small.
- ✓ Personal resources are unavailable for any other method.

Advantage of Direct changeover.

1. It is usually cheap because; two systems are not run in parallel.

Parallel running:

Both the old & the new systems are run side by side (parallel) to each other for some time until users have gained confidence in the new system.

Data is processed on systems in order to compare their performance, and the results are cross-checked. The old system is only abandoned if the new system proves to be satisfactory.

Advantages of parallel running.

1. It promotes user confidence since it allows the results of the old & new system to be compiled side-by-side.
2. It is reliable because it enables thorough testing.
3. The users are given time to familiarize themselves with the new system.

Disadvantages of parallel running.

1. The cost of operating & maintaining the two systems would be high.
2. Resources are duplicated over the two systems, i.e., extra resources have to be engaged to run the two systems in parallel.
3. It is difficult for the staff to carry out clerical operations for two systems during the time available, which is just enough for one system.

Phased changeover:

In phased changeover, the new system is implemented in stages, e.g., one department after the other, while the other departments are still being processed by the old system. When each new phase is proved satisfactory, another one is brought in.

For example;

Changing of the education system from the old curriculum to the new curriculum. Each year, at least one class level changes over to the new syllabus.

Advantage of phased changeover.

1. It ensures slow but sure changeover, and also tends to prolong the implementation period. In this case, users/analysts are able to learn from their mistakes.

Disadvantage of phased changeover.

1. The various elements (i.e., hardware & software) of the same system may be incompatible.

Pilot running:

In pilot running, the overall system is put into use bit-by-bit, e.g., on department basis.

Usually, data from a previous period is first run on the old system and then on the new system.

The results from the new system are then compared with the results from the old. When the new system is considered as correct, a double sequence in one processing run may be tried to convert the pilot run into a parallel run.

Advantage of pilot running.

1. Pilot running offers gradual change to the overall new system. During this time, the staff gains enough experience & confidence.

Disadvantage of pilot running.

1. It is difficult for the staff to carry out clerical operations for two systems.

Security control measures

The data & information in an information system must be kept secure. This is because; if not well protected, the information can be illegally accessed or disclosed to unauthorized parties.

Therefore, the system implementers must make sure that the security features built in the system are properly configured during the implementation stage.

Revision Questions.

1. Define the term 'system implementation'.
2. Explain three tasks that are carried out during system implementation.
3. List two main aims of staff training.
4. Define the following terms:
 - i). File conversion.
 - ii). Changeover.
5. Give an advantage of Straight changeover over Parallel running.

System review

System review is a formal process of going through the specifications, and testing the system after implementation to find out (establish) whether the system meets the original objectives, or whether it performs as predicted in the designed framework.

If the system does not meet/achieve the stated objectives, system development might start all over again.

System review is conducted by the Data processing team of users & auditors. After the review, a post implementation review report is produced, which contains recommendations on how to overcome the problems identified during the review.

System maintenance

Once the system becomes operational, it should be maintained throughout its life.

System maintenance is the process of adjusting & enhancing of requirements, or correcting of errors that may be detected after the system has been implemented in order to keep the system functioning at an acceptable level.

Maintenance functions mainly involve: -

- 1). Correcting errors due to program bugs.
- 2). Changing the procedures & algorithms used to develop the original programs.
- 3). Adding new routines & removing the obsolete routines.
- 4). Hardware & software maintenance.
- 5). Adjusting the existing routines so that the system may adapt to enhanced functional environments.

Reasons for system maintenance.

System maintenance is carried out to ensure that: -

- i). System efficiency is maintained for the changing functional environment & requirements.
- ii). User expectations are satisfied.
- iii). The system caters for growth in application requirements.
- iv). The system adapts to new changes in the company's organization.
- v). System operation is based on the facilities of the current technological development.
- vi). External influences beyond the organization's control are adhered to by the system, e.g., Government policies on taxation, allowances, etc.

NB: System maintenance runs parallel to the maintenance of the system documentation, i.e., any time maintenance is carried out on the system, the documentation should also be updated to convey the right image of the system.

Revision Questions.

1. (a) What is 'system maintenance?'
- (b) Why is system maintenance necessary?

System documentation

Documentation is the process of describing all what the analyst was doing during the system development stages.

Documentation is done by the analyst who developed the system, and is used as a means of communication between the system analyst & the end-users of the system.

NB: System documentation takes place throughout the system development life cycle. After a system has been implemented, any maintenance work must be documented & the analyst modifies or updates the system documents (manuals) so as to reflect the current image of the system.

A comprehensive system documentation consists of the following:

1. Report on fact-finding.
2. Requirement specification.
3. System flowchart and module flowcharts.
4. Table/file structures description.
5. Sample test data & expected output.
6. Output reports.

Report on fact-finding.

At the end of the fact-finding stage, the system analyst should prepare a well detailed report that mainly outlines:

- The methods used to collect data.
- The weaknesses of the current system as indicated by the collected data.
- Recommendations, i.e., why there is need to replace or upgrade the current system.

Requirement specification

The report on requirement specification mainly outlines the:

- Output requirements for the new system, e.g., reports.
- Input requirements.
- Hardware & software required to develop the new system, and also support the processing of an existing system.

System flowchart

The *system flowchart* shows the overall functionality of the proposed information system.

Therefore, at the end of the system design stage, the analyst should write a report that contains:

- The system flowchart or data flow diagrams that shows the processing logic of the information system.
- Any module flowchart that may help programmers in constructing the required subsystem or modules.

Table/file structures description

Depending on the approach used in system construction, the report should contain file or table structure definitions, e.g., if you customised a standard package to construct a system, details on table structures should be well documented.

Sample test data

To test whether the new computerized information system is working as expected, test data is used for every module.

Output reports

To prove that the system is working & giving the desired result, a number of sample outputs from various system modules should be provided.

User manual

User manuals are used to help a person use the system with little or no guidance.

The manual must contain information such as:

1. How to install, start and run the system.
2. How the system appears when running (interface).
3. How to carry out various tasks, e.g., how to include a new entry, data entry, how to modify a record, etc
4. A troubleshooting guide, which describes error correction & how to get help when faced with problems.

Revision Questions.

1. What is 'system documentation?'
2. Outline some of the contents of a user manual.